How systems function

Definition of a system

- A system is an orderly grouping of independent components linked together according to plan to achieve a specific objective.
- The word system is derived from the Greek word “systema” which means the organized relationship among the functioning units.
- However, the word system always comes with an adjective, whenever we talk about a system such as educational system, political system, accounting system etc.

Definition of a system (contd)

- But if we carefully analyse these systems, we can find that there are some features common to all the systems.
- These are the characteristics of the system which help us to understand the working definition of the word system.

Characteristics of a system

The characteristics of the system are

- 1. Basic components
- 2. Interaction and structure
- 3. Goal
- 4. Behavior
- 5. Life cycle

1. System Basic components

- As per the definition of system the functioning units means the basic elements of the system which are interrelated.
- So these basic elements are nothing but the identifiable and moving parts of the system.
- Following are some examples of system and its basic components.

SYSTEM	BASIC COMPONENTS
Educational system	Students, teachers, books, computers
Computer system	Monitor, CPU, keyboard

2. Interaction and structure

- An important feature of the system is the basic components must interact among themselves.
- If an organization, is considered as a system, then for example **purchase department** must interact with **stores and production department**. Therefore, these department are interdependent on each other.
- If we consider, computer as a system then if some information is keyed, it gets processed by arithmetic or logic unit or both and the result is displayed on the screen.
- So, this interrelation activity of the components makes the system dynamic.
- Such a relationship among the components which define the boundary between the system and environment is called as the structure of the system.

3. Goal

- In order to achieve the goal of the system we should first understand the meaning of **Central objective, Integration,** and **Synergistic effect**

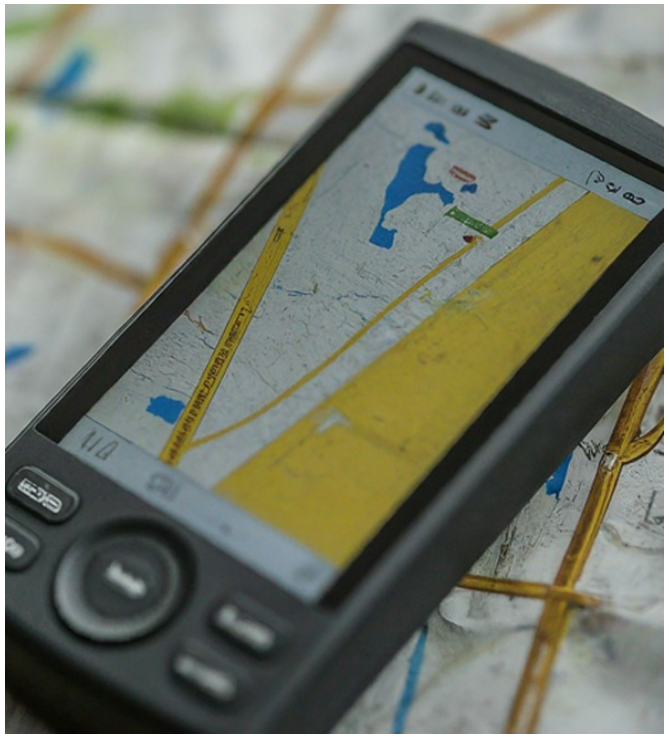
1. **Central Objective** : means the common goal, because without common goal system will start moving in all directions. As a result, coordination among all the parts (components) will be lost.
2. **II. Integration:** It is combined work of all the components in order to achieve the goal of the system. There must be coordination among all parts of the system So in order to have such coordination the system must work as a' whole', integrating all its activities to achieve the desired result.

3. Goal (contd)

3. **Synergistic effect:** From the integration concept it is clear that the system has to be viewed as ‘whole’ rather than just as sum of its parts. This integrating effect is called as synergistic effect.

4. Behaviour

- Behavior is the way the system reacts to its surrounding environment.
- Behavior is determined by the procedures designed to make sure that components behave in ways that will allow system to achieve common goal. E.g.
  - ✓ For example, when you use a GPS navigation app on your smartphone to find the nearest restaurant, the app acts as the interface between you and the GPS system.
  - ✓ The app collects your current location from the GPS receiver, processes it with map data, and provides directions to the restaurant.
  - ✓ This involves input, processing, output, and control components of an information system.



4. Behaviour (contd)

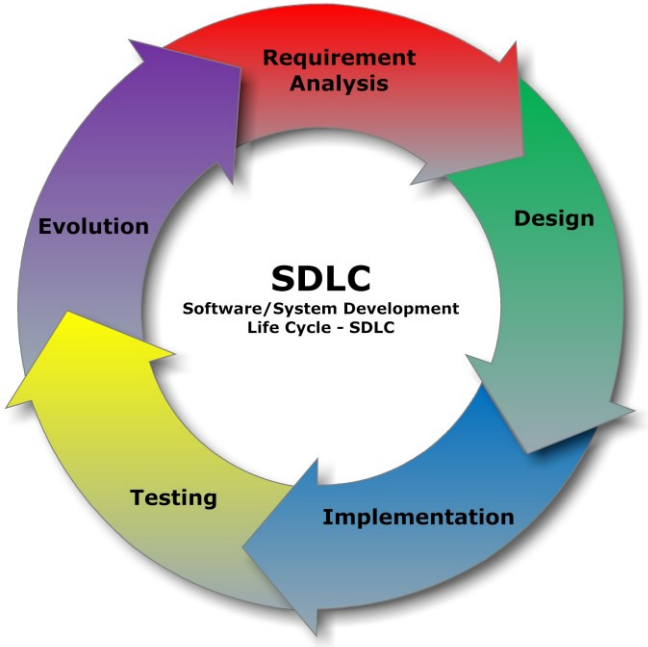
- **Input**: GPS receivers constantly collect signals from multiple satellites orbiting the Earth.
- **Process**: The GPS receiver calculates its position, speed, and time based on the received signals. This involves complex mathematical calculations.
- **Output**: The processed information is displayed on a device screen as coordinates, maps, or directions.
- **Storage**: GPS devices often store location history, waypoints, and maps for offline use.
- **Control**: The GPS system's software manages the entire process, from signal acquisition to data processing and output.

4. Behaviour (contd)

- So a signal is input from environment, reaction is the behavior and instruction in the GPS system (how to react) is the procedure.
- Procedure describes what ought to be done and behavior describes what is actually done.

5. Lifecycle

- Every system has life cycle and according to human life it has birth that is evolution, life, aging, repairs and finally the end of the existence of the system (death).



Elements of the system

- All the characteristics of the system are determined by the system elements, their properties and relationships. The system elements are: Input, Processor and Output
- These elements are common to all systems.
- These are the elements by which all systems are described.
- They are set in a fixed position which helps the system analyst to design and work with system more easily.

Elements of the system (contd)

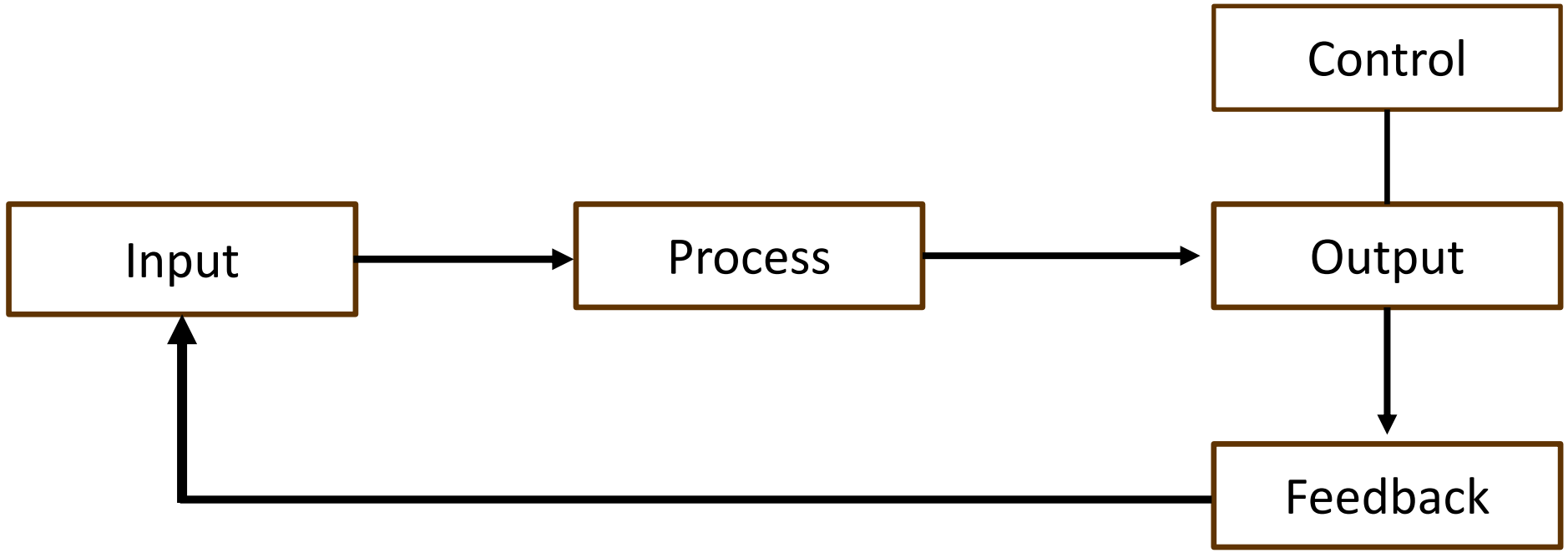
1. **Input**
   - It is defined as energizing or start up component on which system operates.
   - It may be raw material, data, physical source, knowledge or any energy to decide the nature of output.
2. **Processor**
   - It is defined as the activity that makes possible the transformation of input to output.
   - When data is processed through computer it is processed through logical steps.
   - However, these steps are required to be instructed in series to the computer.

Elements of the system (contd)

3. **Output**

- It is the result of the operation. In other words, it is the purpose or the main objective for which the system is designed.
- Though output is largely dependent on input, its nature or format may vary vastly from the input. For example: If data keyed is in numerical form it may display output which is in form of graph or pictorial form.

Information system environment



Information system Environment (contd)

- In order to build any system only the knowledge of its elements does not serve the purpose, there should be fundamental clarity of some important concepts which are essential to build the efficient system and to keep it in equilibrium.
- The major concepts are:
  1. Boundary & environment
  2. Subsystem
  3. Interface
  4. Feedback control
  5. Black box

Information system Environment (contd)

1. **Boundary and environment**

- Every system has its limits that determine the sphere of influence and control is called as Boundary of the system.
- Everything within the circumscribed space is called system and everything outside it is environment.
- Flow from environment to the system is its input while a flow from system to its environment is the output.
- Boundary of the system may exist physically or conceptually.



Information system Environment (contd)

2. Subsystem

- A complex system is difficult to implement when considered as a whole.
- However, if we divide it into smaller functional units which are of manageable sizes then every small function unit becomes a subsystem.
- In the formation of subsystem, the components performing same or similar functions are grouped, for example: in a business organisation system, marketing, production, sales can be considered as subsystems.
  - ✓ **Module**: A collection of function or data. In other words, module encapsulates related functions. Ideal module is that module which can be reused in other development projects.

Information system Environment (contd)

3. **Interface**

- The interconnections and interactions among the subsystems are termed as interfaces.
- In fact, each interface implies a communication path. Number of interfaces increase with number of subsystems.

Information system Environment (contd)

4. **Feedback control**

   - In order to improve the performance of any system feedback control mechanism can be used as a tool or device to control or modify the input of the system after analyzing the output properly.

Information system Environment (contd)

5. Black box

- Black box is the subsystems at lowest level where the inputs are defined, outputs are determined but the processor of the system is not defined
- This means it is difficult to understand how the transformation of input to output takes place.

